THE DEGREE PAIRING OF THE COMPONENTS OF THE SPECIAL FIBRE OF A REGULAR MODEL IS SYMMETRIC

THE STACKS PROJECT AUTHORS

Editorial note. The counted unit is the Stacks Project Lemma lemma-intersection-pairing at source lines 4697-4711 together with its proof at source lines 4713-4734, a straight adjacent pairing in the chapter file. The chapter is the upstream file models.tex, which is the file name recorded in the source record; the pool directory carries the longer name stacks-models and is not used as the chapter name anywhere in this card.

Editorial note. The retained statement opens with the words In Situation followed by a reference to situation-regular-model, so the situation environment that defines the setting is retained as a context span at source lines 4628-4637. It is context, not a counted unit, and the interval of the counted lemma does not overlap it.

Editorial note. This chapter's candidate index records the statement body and the proof body without their closing markers, so the retained spans used here are the full environments, exactly as in the earlier cards of this chapter.

Editorial note. The proof quotes two lemmas of the Varieties chapter and the retained situation quotes one lemma of this chapter; all three are declared as non-load-bearing navigation pointers with their stable Stacks tags looked up in the shared tag index. No citation occurs in the retained spans, so no bibliography entry is supplied.

Editorial note. The retained preamble is the chapter preamble's Xy-pic setup at lines 14-18, the theorem declarations and mathematical macros at lines 154-217, and the three no-op wrapper environments.

Allowed standard prerequisites (editorial).

- The regular-model situation and the degree of an invertible sheaf on a component. The situation fixes a discrete valuation ring with fraction field, residue field and uniformizer, a smooth projective curve over the fraction field, a regular proper model of it, and the irreducible components of the special fibre with their multiplicities; the degree of an invertible sheaf restricted to a component is the degree of the corresponding divisor on the proper curve over the residue field. The author's own situation and the two degree lemmas of the Varieties chapter that the proof quotes are the allowed inputs.. The counted lemma builds the pairing that the rest of the section uses to state the symmetry of the intersection numbers of the components, and the statement itself is phrased inside that situation.

Source reference. Original lemma 0C5Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15279–15279. Navigation pointer inside the retained situation, which names it when it writes the special fibre as a sum of its components with multiplicities; the counted lemma is the symmetry of the pairing and does not depend on the multiplicities. The number is the stable Stacks tag of the target, looked up in the shared tag index.

Date: stacks-project:models.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AYX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13805–13805. Allowed input of the Varieties chapter quoted by the retained proof for the additivity of the degree under tensor products; the pointer is registered so the reference resolves and the cited statement is not reproduced inside the span. The number is the stable Stacks tag of the target, looked up in the shared tag index.

Source reference. Original lemma 0AYY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13806–13806. Allowed input of the Varieties chapter quoted by the retained proof for the degree of an effective Cartier divisor on a curve; the pointer is registered so the reference resolves and the cited statement is not reproduced inside the span. The number is the stable Stacks tag of the target, looked up in the shared tag index.

Situation 0.1. Let R be a discrete valuation ring with fraction field K, residue field k, and uniformizer π. Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$. Let X be a regular proper model of C. Let $C_1, \ldots, C_n$ be the irreducible components of the special fibre X_k. Write $X_k = \sum m_i C_i$ as in Lemma 0C5Z.

Lemma 0.2. *In Situation 0.1 given $\mathcal{L}$ an invertible $\mathcal{O}_X$-module and $a = (a_1, \ldots, a_n) \in \mathbf{Z}^{\oplus n}$ we define*

$$\langle a, \mathcal{L} \rangle = \sum a_i \deg(\mathcal{L}|_{C_i})$$

Then $\langle , \rangle$ is bilinear and for $b = (b_1, \ldots, b_n) \in \mathbf{Z}^{\oplus n}$ we have

$$\Big\langle a, \mathcal{O}_X(\sum b_i C_i) \Big\rangle = \Big\langle b, \mathcal{O}_X(\sum a_i C_i) \Big\rangle$$

Proof. Bilinearity is immediate from the definition and Varieties, Lemma 0AYX. To prove symmetry it suffices to assume a and b are standard basis vectors in $\mathbf{Z}^{\oplus n}$. Hence it suffices to prove that

$$\deg(\mathcal{O}_X(C_j)|_{C_i}) = \deg(\mathcal{O}_X(C_i)|_{C_j})$$

for all $1 \leq i, j \leq n$. If $i = j$ there is nothing to prove. If $i \neq j$, then the canonical section 1 of $\mathcal{O}_X(C_j)$ restricts to a nonzero (hence regular) section of $\mathcal{O}_X(C_j)|_{C_i}$ whose zero scheme is exactly $C_i \cap C_j$ (scheme theoretic intersection). In other words, $C_i \cap C_j$ is an effective Cartier divisor on C_i and

$$\deg(\mathcal{O}_X(C_j)|_{C_i}) = \deg(C_i \cap C_j)$$

by Varieties, Lemma 0AYY. By symmetry we obtain the same (!) formula for the other side and the proof is complete. □